Introduction to Artificial Intelligence

Prepared by: Gopi Prajapati

What is Intelligence?

- The ability to reason
- The ability to understand
- The ability to create
- The ability to learn from experience • The ability to plan and execute complex tasks

Artificial Intelligence

- "Giving machines ability to perform tasks normally associated with human intelligence." • Branch of computer science that aims to create intelligence of machines
- Part of computer science concerned with designing intelligent machines

Continue..

- Deals with issues like inference, reasoning, problem solving, knowledge representation, planning, natural language processing, perceptron, etc.

- Can be defined by two dimensions:

System that thinks like humans	**System that think rationally**
System that acts like humans	System that act rationally

Acting Humanly: The Turing test approach

- proposed by Alan Turing (1950)
- designed to convince the people that whether a particular machine can think or not • a test based on indistinguishability from undeniably

intelligent entities- human beings • Involves an interrogator who interacts with one human and one machine. Within the given time the interrogator has to find out which one is human and which one is machine

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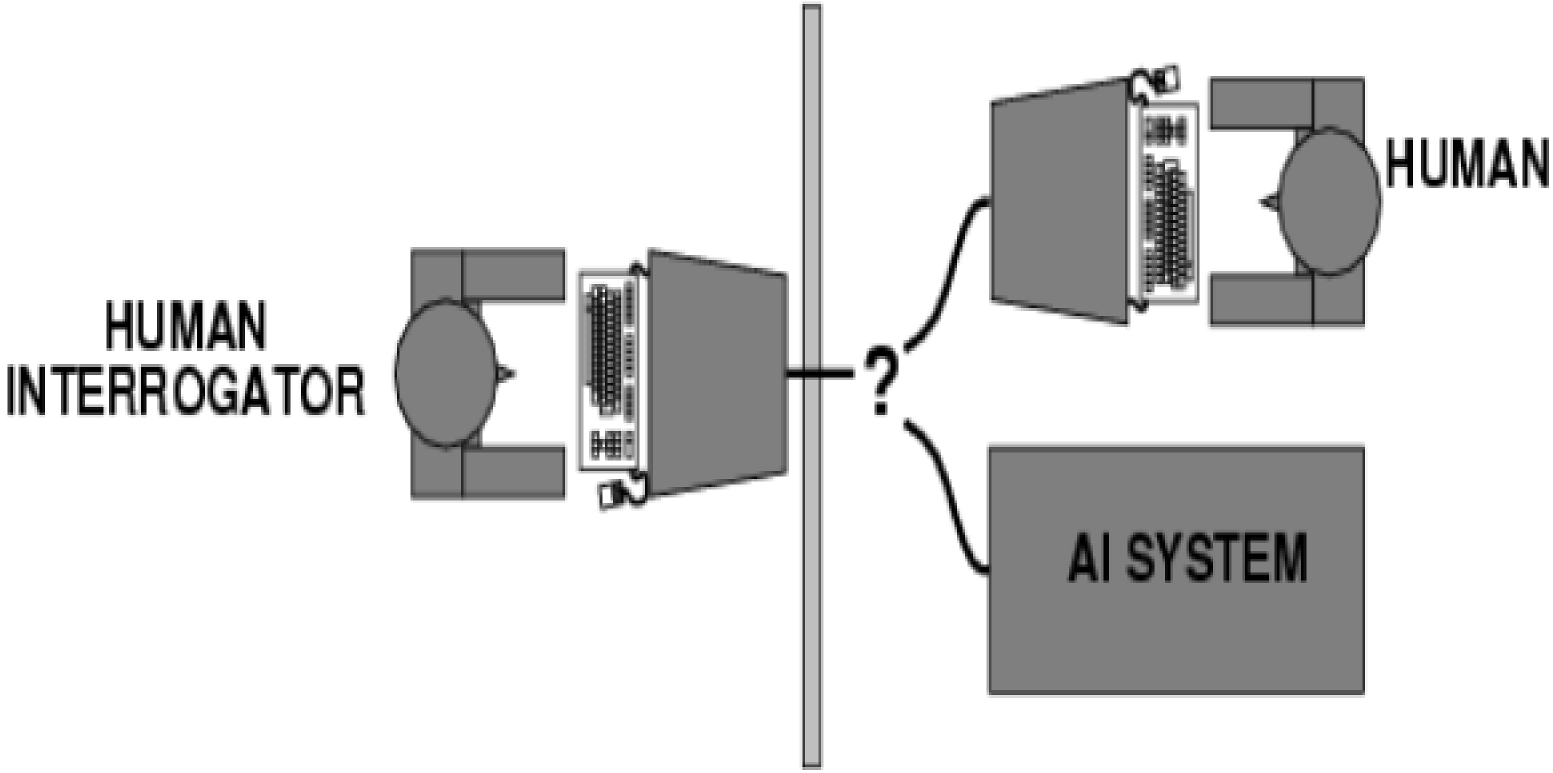

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- The computer passes the test if a human interrogator can not tell whether the written response come from human or not

- To pass a Turing test, a computer must have following capabilities:

- Natural language processing
- Knowledge representation
- Automated reasoning
- Machine learning

Chinese room argument

- Searle (1999) summarized the Chinese Room argument
- Imagine a native English speaker who knows no Chinese locked in a room full of boxes of Chinese

symbols (a data base) together with a book of instructions for manipulating the symbols (the program).

- Imagine that people outside the room send in other Chinese symbols which, unknown to the person in the room, are questions in Chinese (the input).

Chinese room argument

- And imagine that by following the instructions in the program the man in the room is able to pass out Chinese symbols which are correct answers to the questions (the output).

- The program enables the person in the room to pass the Turing Test for understanding Chinese but he does not understand a word of Chinese.

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- Turing test avoid the physical interaction with human interrogator.
- The total Turing test computer must have following additional capabilities:

Computer Vision: To perceive objects

Robotics: To manipulate objects and move

Thinking Humanly: Cognitive approach

- Make the machines with mind.
- Cognition means the action or process of acquiring knowledge and understanding through thought, experience and senses. • Study of Mind-Body nature
- Not possible to make machines that think like human brain

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- To make a machine that think like human brain, cognitive model is required. Two ways of doing this is:

– Predicting and testing human behavior (cognitive science)

– Identification from neurological data (Cognitive neuroscience)

Thinking rationally: The laws of thought approach

- Aristotle was one of the first who attempted to codify the right thinking.
- He gave Syllogisms that always yielded correct conclusion when correct premises are given.
-

For example:

- Ram is a man
- Man is mortal

=>Ram is mortal

Continue..

- To create intelligent systems that give correct result using logic programming
- Problems: It is not easy to take informal knowledge and state in the formal terms required by logical notation, particularly when knowledge is not 100% certain. (like

theoretical and practical)

Acting Rationally: The rational Agent approach

- Agent is something that acts. Computer agent is expected to have following attributes: ▪ Autonomous control

▪ Perceiving their environment

▪ Persisting over a prolonged period of time

▪ Adapting to change

▪ And capable of taking on another’s goal

Continue..

- Rational Agent is one that acts to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- Rational behavior means doing the right thing.
- The right thing is that which is expected to maximize goal achievement, given the available information

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- to act rationally is to reason logically to the conclusion and act on that conclusion • It is more general than laws of thought approach

- It is more controllable than approaches based on human behavior or human thought because the standard of rationality is clearly defined and completely general.

Applications of AI

- Game playing
- Speech recognition
- Pattern recognition
- Understanding natural language
- Computer vision

- Expert systems
- Heuristic classification
- Data mining

Foundations of AI

- Philosophy
- Mathematics
- Psychology
- Economics
- Linguistics
- Neuroscience
- Sociology

- Politics

Goals of AI

- Problem solving and decision making
- Automation
- Learning and adaptation
- NLP
- Perception and vision
- Human AI collaboration
- Reasoning and understanding
- Ethical and responsible AI
- Achieving artificial general intelligence

Challenges of AI

- AI ethical issues
- Bias In AI
- AI integration
- Computing power
- Data privacy and security
- Legal issues
- AI transparency
- Limited knowledge
- Building trust
- Lack of AI explain ability

Brief history of AI

1. The Early Foundations (1940s–1950s)

- **1943:** Warren McCulloch and Walter Pitts proposed a model of artificial neurons, laying groundwork for neural networks.

- **1950:** Alan Turing published *Computing Machinery and Intelligence*, introducing the **Turing Test** to evaluate machine intelligence.
- **1956:** The term **"Artificial Intelligence"** was coined at the Dartmouth Conference, organized by John McCarthy, Marvin Minsky, Claude Shannon, and others—considered the birth of AI as a field.

2. Early Optimism and First AI Programs (1950s–1960s) • Programs were developed to solve logic puzzles, play checkers, and prove theorems.

- **1956:** Logic Theorist (by Newell, Shaw, and Simon) could prove mathematical theorems.
- **1959:** Arthur Samuel's checkers program learned from experience, pioneering machine learning.
- **1960s:** Early natural language processing (ELIZA, 1966) and robotics

(Shakey the robot) emerged.

3. AI Winters and Challenges (1970s–1980s)

- Progress slowed due to limited computing power, data, and unrealistic expectations.
- **First AI Winter (1974–1980):** Reduced funding after critical reports (e.g., Lighthill Report, 1973).
- **Expert Systems** gained popularity in the 1980s—programs that mimicked human expertise in narrow domains (e.g., MYCIN for medical diagnosis).
- **Second AI Winter (late 1980s):** Expert systems proved expensive and limited, leading to another downturn.

4. Revival and Rise of Machine Learning (1990s–2000s)

- **Machine learning** shifted focus from rule-based systems to data-driven approaches.

• **1997:** IBM's **Deep Blue** defeated world chess champion Garry Kasparov. • **2000s:** Increased computational power, big data, and improved algorithms enabled practical applications (search engines, recommendation systems). • **2006:** Geoffrey Hinton's work revitalized **deep learning** using neural networks with many layers.

5. Modern AI Boom (2010s–Present)

- **2012:** AlexNet dramatically improved image recognition accuracy, sparking the deep learning revolution.
- **2014–2017:** Breakthroughs in **generative adversarial networks (GANs)**, **transformers**, and **reinforcement learning**.
- **2016:** Google DeepMind's **AlphaGo** defeated world champion Lee Sedol in Go.
- **Late 2010s–2020s:** Rise of large language models (GPT series, BERT), AI in everyday tech (voice assistants, autonomous driving), and generative AI (DALL·E, ChatGPT).

Omniscience

- Omniscience is the capacity to know everything including the future
- the capacity to have unlimited knowledge

about all things.

- Knowing everything means that for any possible situation or question, you already have the solution.
- There is no need to involve logic or rational thought.

Aspect	Artificial Intelligence (AI)	Human Intelligence (HI)
Definition	Intelligence demonstrated by machines through data processing and algorithmic learning.	The innate cognitive ability of humans to think, reason and adapt based on experience and emotions.

Nature	Simulative – mimics human actions using computational logic.	Adaptive – integrates emotion, experience and reasoning.
Structure	Operates through neural networks, algorithms and digital systems.	Operates through biological neurons and cognitive processes.
Learning Method	Learns through data, feedback loops and training datasets.	Learns through experiences, environment and social interaction.
Decision Making	Objective and data-driven; lacks ethical or emotional context.	Subjective; influenced by logic, emotions and moral considerations.

Creativity	Limited to programmed boundaries but lacks imagination.	Capable of innovation, abstract thinking and creative expression.
Adaptability	Adapts only when re-trained with new data.	Adapts naturally to changing environments and situations.
Speed & Efficiency	Processes data at high speed with minimal error.	Slower in computation but excels in contextual reasoning.
Error Rate	Low — depends on data quality and algorithm accuracy.	Higher — influenced by human bias, fatigue or emotional state.

Ethics & Morality	Lacks ethical awareness or moral sense.	Guided by conscience, empathy and ethical judgment.
Social Interaction	Limited understanding of emotions and social cues.	Highly capable of emotional intelligence and interpersonal communication.
Multitasking	Performs specific tasks efficiently, often limited to one function.	Can multitask, think critically and switch contexts fluidly.
Physical Limitation	Can operate continuously without fatigue.	Needs rest, sleep and nutrition to function effectively.

Agents and Problem solving Prepared by: Gopi Prajapati

Agent

- An agent is anything that perceives its environment through sensors and acts upon that environment through sensors.
- Its action can change environment through different states.

- The sensors act as input device and actuators act as output device.
- The percept is complete set of inputs at a given time.

Agent

- It has agent program that decides the best action to be taken based upon current percept.
- The mapping between percept and its action is done by function.
- Agent function can be in terms of table or

rule.

$$f : P^* \rightarrow A$$

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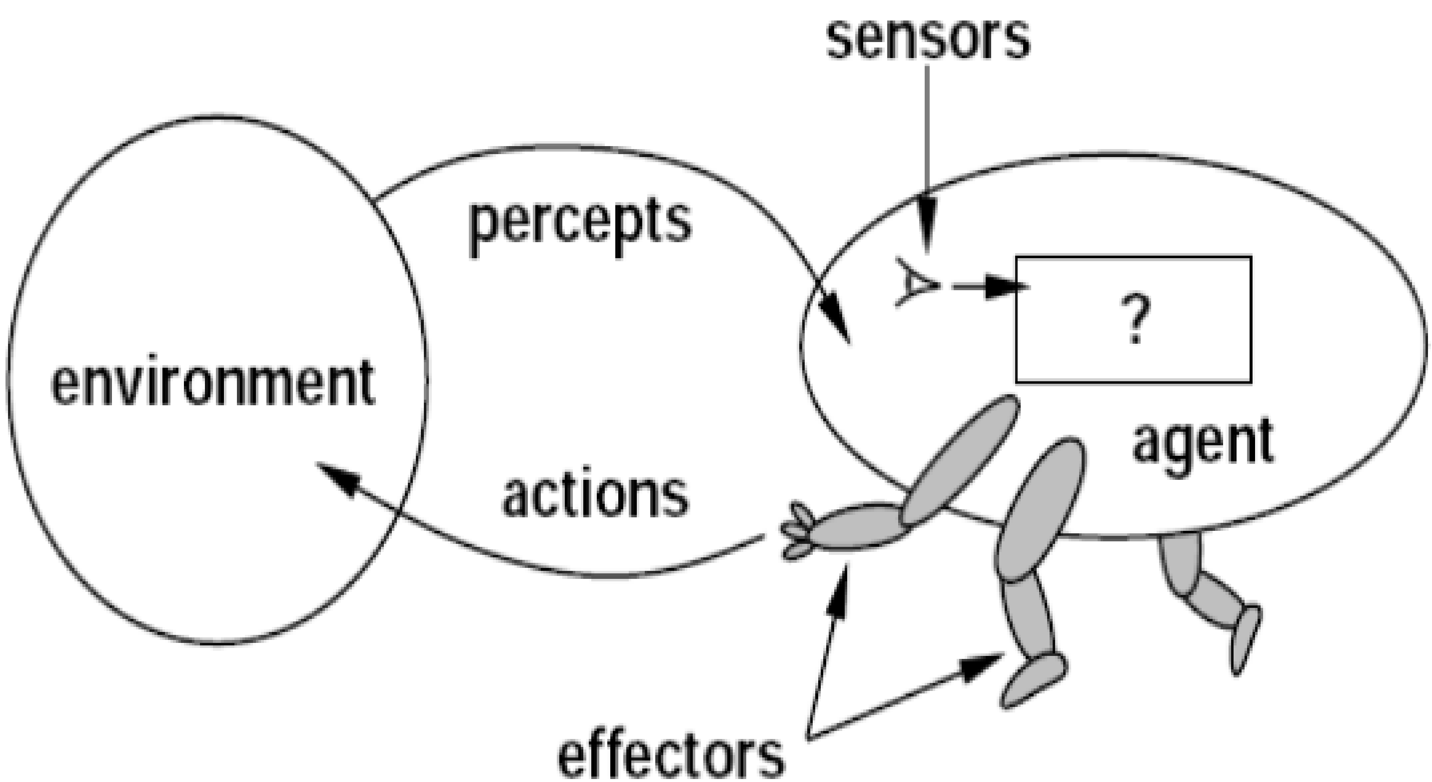


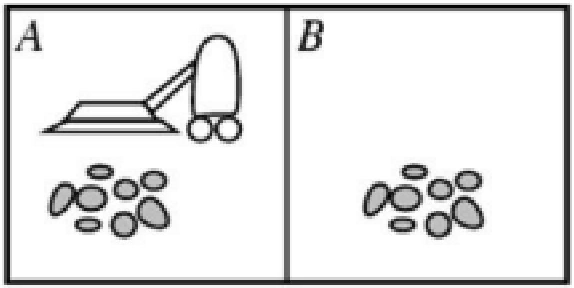
Sensors, Percepts, Effectors and Actions

- For human,
- Sensors: eyes, skin, tongue, nose, ears • Percepts: Signals from the sensors, location, texture, color, direction
- Effectors: legs, hands, eyes, head
- Actions: lift finger, walk, talk, run

Example of agent

Vacuum-cleaner world

- **Percepts:**
  Location and status,
  e.g., [A,Dirty]
- **Actions:**
  Left, Right, Suck, NoOp



Example vacuum agent program:

function Vacuum-Agent([location,status]) returns an action

- *if* status = Dirty *then* return Suck
- *else if* location = A *then* return Right
- *else if* location = B *then* return Left

Properties of Agents

- Autonomous

- Interacts with other agents and the environment
- Reactive to the environment
- Pro active/ goal directed

Types of agent

a. Intelligent agent: This type of agent must sense, act, be autonomous and be rational.

b. Rational agent: Rational agent always does right thing based upon its prior knowledge, percept history, its ability and its constraints like time and space limitation. A rational agent can be perfectly rational without possessing the ability to learn and be autonomous. It always act to maximize its expected value.

All rational agents are not necessarily intelligent but all truly intelligent agents are expected to be rational within their operational domain.

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c. Omniscient Agent:

This type of agent knows the actual outcome of its action and can act accordingly. But omniscience is impossible in reality.

Agent Structure: Table based agent

In this agent, the mapping between percepts and its action is stored in the form of table

- it looks on table and takes the appropriate actions
- It is good for simple task or environment where agent’s action depends upon only current percept.
- It has no autonomous and no learning.

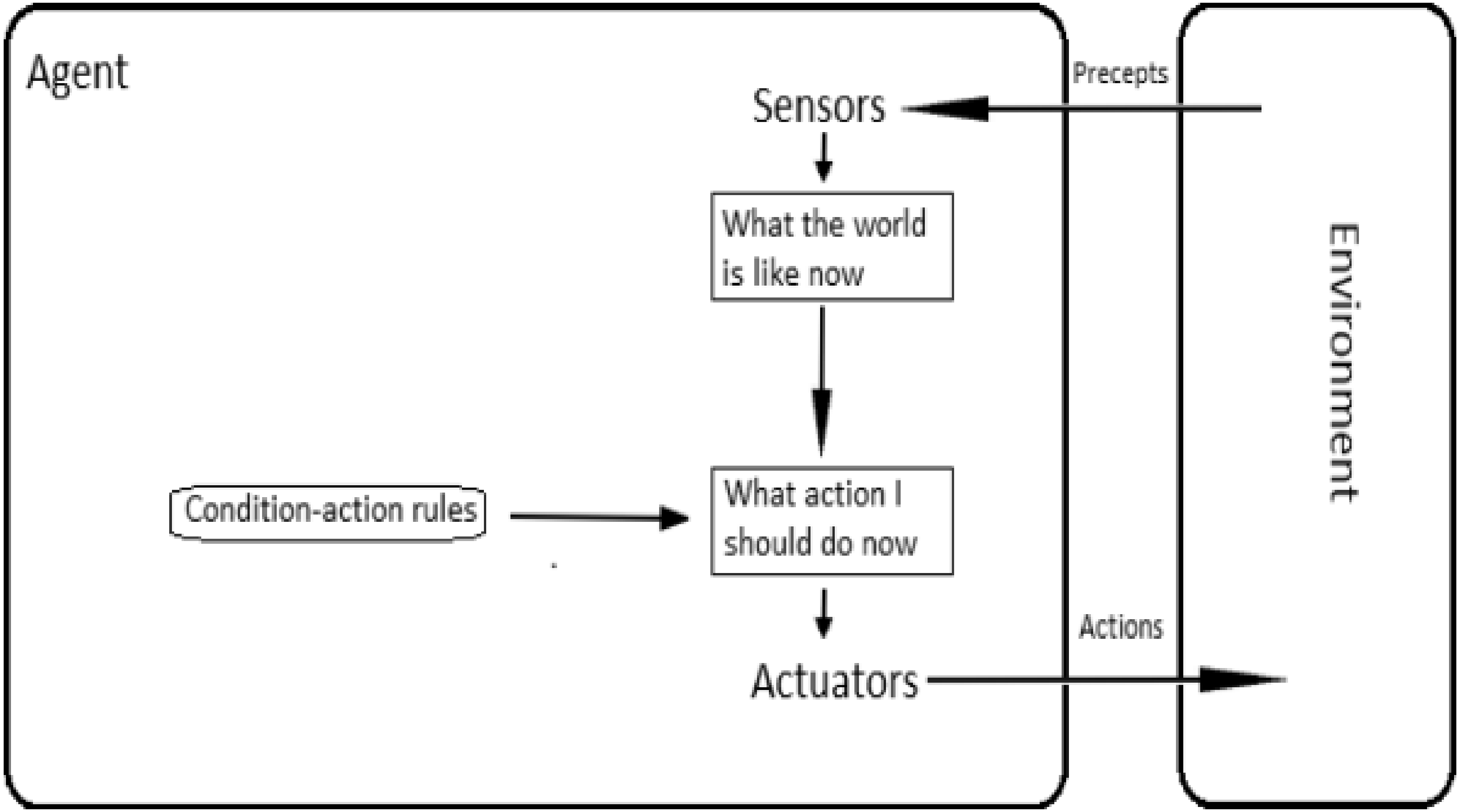
Example

- Vacuum cleaner

[A, Dirt]	Suck
[A, Clean]	Move right

Simple Reflex Agent

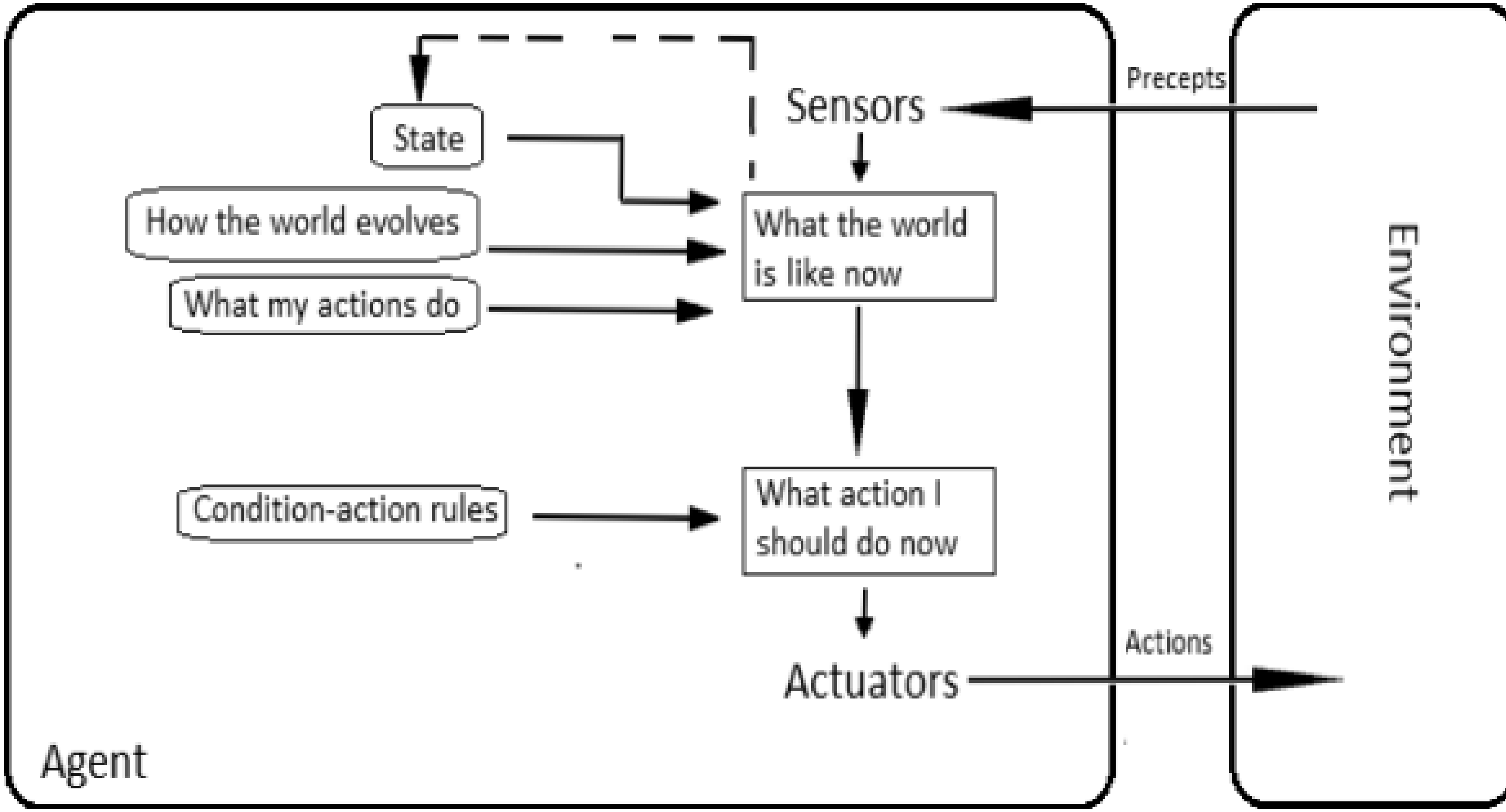
- It responds immediately to the percept by matching the percept into action.
- Such matching is done by using some rules or policy.
- Such type of agent is very simple and good for fully observable environment.
- It lacks learning.
- For e.g. Whenever I touch the heater, I immediately remove my hand from it. It means it can’t show deliberate action as it does not think about situation it has perceived.



Model Based agent

• This type of agent has some internal memory • It keeps world model in some memory to remember what was perceived before and keep track the different aspects of the world. • This allows the agent to compute the next state of the world based upon the current state and world model.

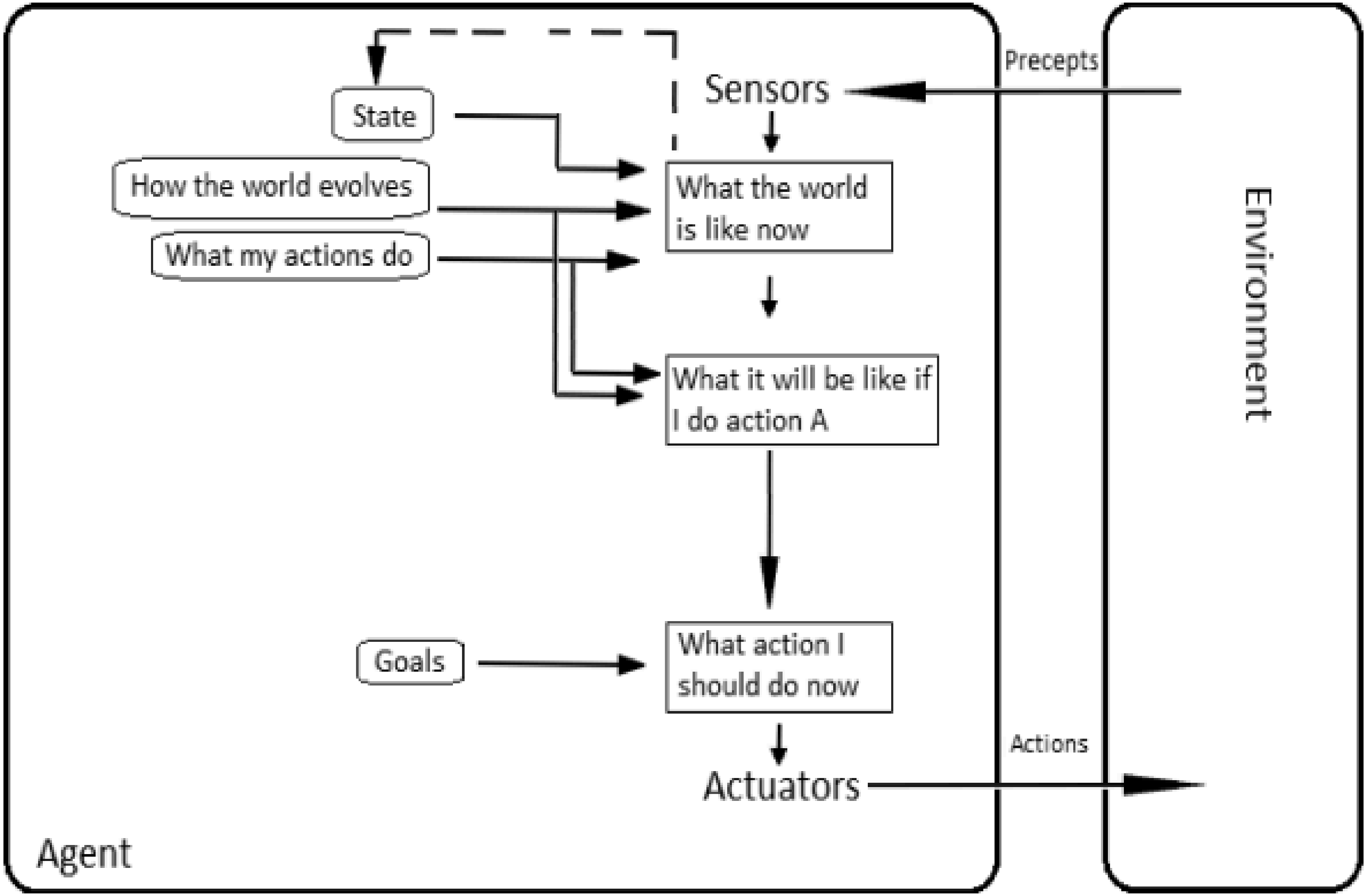
- This type of agent is good for partially observable environment. It is good for games like puzzle, chess etc.



Goal Based Agent

- This type of agent has some goal to achieve with its world model, perception history and list of actions it can perform.

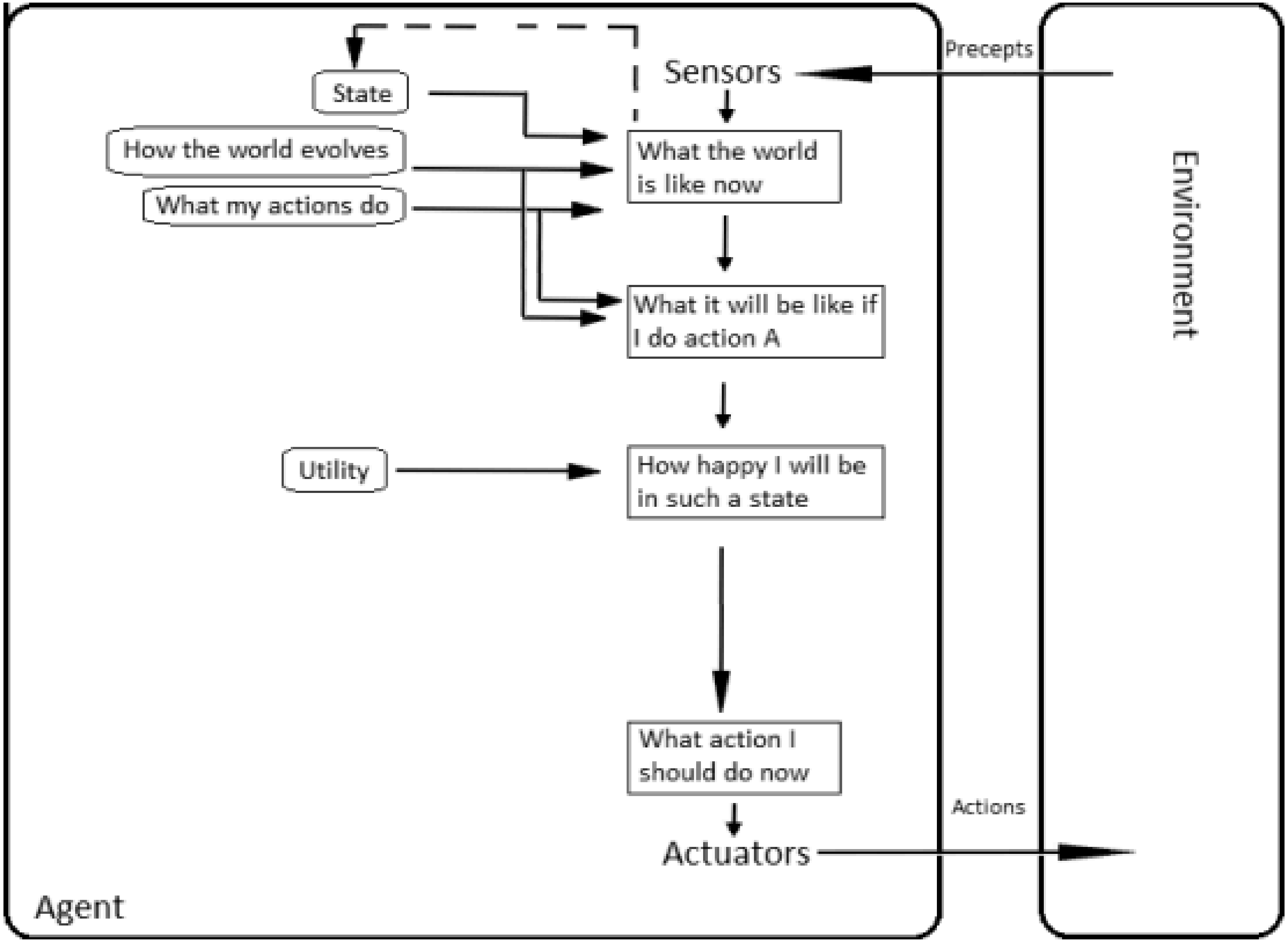
- It tries sequence of actions that can lead to goal internally.
- If it does, it takes that action. i.e. It check whether this action takes me to the goal or not.
  - It can analyze and think over current situation.



Utility Based Agent

- It is a goal based agent that tries to maximize its utility.

- it tries to achieve optimal goal.
- Some goals can be achieved different ways and every way has some preferences or utility. • chooses the action that maximizes the expected utility of the action outcomes
- has to model and keep track of its environment, tasks, perception, representation, reasoning, and learning.



Learning agents

- A learning agent learns about environment model
- learns about its action consequences and its utility.
- It learns with the feedback from environment.
- Learning enables an agent to adapt new

situations without being reprogrammed by humans.

Performance Evaluation of agents

- Performance measure tells how successful an agent is.

- Agent changes the environment into different sequence of states.
- If the states are desirable then, agent has performed well.

PEAS Description

- To design an agent we must specify its task environment. Task environment includes:

- Performance
- Environment
- Actuators and
- Sensors

- It provides a comprehensive view of how AI agents function.

- Each element of PEAS defines a specific aspect of the agent’s operation, guiding how the AI interacts with its environment, makes decisions and executes actions to achieve its objectives.

PEAS description of Fully automated taxi:

- **P - Performance Measure**: How success is judged.
  - Safety (no accidents)
  - Efficiency (minimizing travel time, fuel)
  - Passenger Comfort (smooth ride, appropriate speed)
  - Obeying Traffic Laws
- **E - Environment**: The surroundings the car operates in.
  - Roads (lanes, intersections, construction)
  - Traffic (other cars, pedestrians, cyclists)
  - Weather (rain, fog, sun glare)
  - Traffic Signals & Signs
- **A - Actuators**: The physical mechanisms the car uses to act.
  - Steering Wheel (to turn)
  - Accelerator & Brakes (to control speed)
  - Turn Signals, Horn, Wipers, Lights
- **S - Sensors**: The inputs the car receives from the environment.

– Cameras (for object/lane detection)
– LiDAR & Radar (for distance & mapping)
– GPS (for navigation)
– Accelerometers & Odometers (for speed/distance

PEAS description for vacuum cleaner

- **P - Performance Measure**

i. Maximize cleanliness, minimize time, reduce energy use, ensure safety, avoid obstacles.

ii. Clean all dirty spots, return to base, complete tasks efficiently. •

E - Environment

i. A bounded space (e.g., a house with rooms, furniture, carpets). ii. Contains clean and dirty locations, obstacles (tables, wires), varying floor types.

- **A - Actuators (How it acts)**

i. Movement: Move forward, turn left/right, stop. ii. Cleaning: Suck/clean (remove dirt).

iii. Control: Navigate, return to charging station, empty bin. •

S - Sensors (How it perceives)

i. Dirt Sensor: Detects if the current tile/area is dirty. ii. Location Sensor/Map: Knows its position and navigates. iii. Obstacle Sensors:

Bump sensors, cliff sensors, laser rangefinders (LIDAR).

iv. Camera/Vision: For mapping and obstacle avoidance

PEAS for medical diagonosis system

• **P - Performance Measure (How success is judged)**

i. Correctness & Accuracy: High diagnostic accuracy, fewer misdiagnoses. ii. Efficiency: Speed of diagnosis, reduced time to treatment.

iii. Patient Outcome: Improved health, reduced risk, cost-effectiveness. iv. Completeness: Thoroughness of recommendations (tests, referrals). • **E - Environment (Where the agent operates)**

i. Clinical Settings: Hospitals, clinics, remote patient monitoring. ii. Data Sources: Patient records, medical literature, lab results. iii. Interacting Entities: Patients, doctors, nurses, insurers, courts. iv. Dynamic Factors: Evolving patient conditions, new medical knowledge. • **A - Actuators (Actions the agent can take)**

i. Output: Displaying diagnoses, treatment plans, risk assessments on screens. ii. Recommendations: Suggesting further tests, specialist referrals, medication. iii. Data Generation: Triggering scans, lab orders.

• **S - Sensors (Inputs the agent perceives)**

i. Patient Data: Symptoms (reported, observed), vital signs, medical history, genetic data.

ii. Medical Records: Digital patient files, past treatments, test results (X-rays, blood work).

iii. External Data: Real-time IoT medical device feeds, research databases

Characteristics of Task environment

1. Fully observable Vs partially observable: • In fully observable environment, agent's sensor can access to the complete state of environment at each point. For eg. chess • The relevant features may be partially observable in some cases. For eg. Poker

2. Deterministic Vs Stochastic:

- In deterministic environment, the change in environment is known. For eg. Chess. • In stochastic, the change in environment is uncertain. For eg. Ludo

3. Episodic Vs sequential:

- Certain task can be divided into different episodes. The action of one episode does not affect subsequent episodes. For eg. determining defective parts.
- If current action may affect other episodes then, it is called sequential. For eg. Playing chess, taxi driving.

4. Discrete Vs Continuous:

- If number of actions, percepts and states are limited, it is discrete environment. For eg. Chess playing

- If infinite then continuous. For eg. taxi driving

5. Single Agent Vs Multi agent:

- Environment having single agent that operates itself is called single agent environment. For eg. crossword puzzle solving agent.
- If it is two or more agent, then it is multi agent. For eg. chess playing.

Problem as state space

- A problem is defined by:

– An initial state

– Successor function: Description of possible actions – Goal test: Determine whether the given state is goal state or not

– Path cost: Sum of cost of each path

- A solution is a sequence of actions from initial to goal state. Optimal solution has the lowest path cost.
- Approach to problem solving through a range of possible states in order to reach Some predefined goal
- To provide with optimal cost solution

State Space: The **set of all possible states** reachable from the initial state through any sequence of actions.

- Can be finite or infinite.
- Often represented as a **graph** where nodes are states and edges are transitions.

Four general steps in problem solving

– Goal formulation

– Problem formulation

– Search

– Execute

Measuring problem Solving Performance

- **Completeness**

– An algorithm is said to be complete if it definitely finds solution to the problem, if exist.

- **Time Complexity**

– How long (worst or average case) does it take to find a solution?

- **Space Complexity**

– How much space is used by the algorithm?

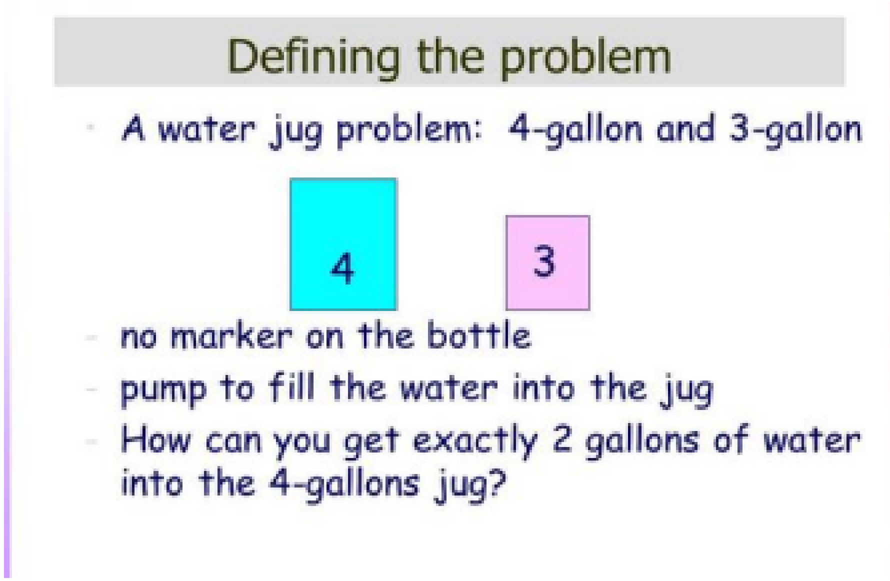
- **Optimality/Admissibility**

– Does it give optimal solution?

The water jug problem

Problem: There are two jugs, a 4-gallon one and a 3-gallon one. Neither jug has any measuring markers on it. There is a pump that can be used to fill the jugs with water.

- How can you get exactly n (0, 1, 2, 3, 4) gallons of water into one of the two jugs ?



Here rules applied are:

2. Fill the 3 gallon jug

9. Pour all the water from 3 gallon jug to 4 gallon jug

2. Fill the 3 gallon jug

7. Pour the water from 3 gallon jug into 4 gallon jug until 4 gallon jug is full

Problem Types

There are several categories of problems that determine which search methods and agent types are appropriate:

- **Single-State vs. Multiple-State Problems**

Single-State: Agent knows exactly where it is. The solution depends only on the current state's properties, with no need to consider future states or transitions.

Characteristics: Simple, static environments, straightforward solutions, limited scope.

Example: A simple logic problem where you check if a single condition is met (e.g., "Is the number even?")

Multiple-State: Agent is unsure and must consider multiple possibilities. Requires exploring a series of states and actions (a state space) to reach a goal, often involving uncertainty or limited knowledge of the environment.

Characteristics: Complex dynamics, need for path finding, potential for incomplete information (inaccessible states).

Examples:

- **Navigation**: Finding a route from City A to City B on a map (states are locations, actions are travel).

- **Observable vs. Hidden Information**

Observable: All relevant information is accessible. In the weather example, a friend's daily activity (walking, shopping, cleaning) serves as an observable signal that depends on the hidden weather.

Hidden: Some parts of the state are unknown. For example, the actual weather (sunny, cloudy, or rainy) is a hidden state that you cannot directly observe if you are indoors.

- **Static vs. Dynamic Environments**

Static: Environment remains unchanged while solving. The environment doesn't change while the agent is deciding on an action; time is irrelevant to the environment's state.

Characteristics: Complex, unpredictable, requires real-time processing and adaptation.

Examples: Traffic, pedestrians, and weather are always in motion.

Dynamic: Environment can change unpredictably. The environment changes while the agent is deliberating or acting, requiring constant monitoring.
Characteristics: Complex, unpredictable, requires real-time processing and adaptation.

Examples: Traffic, pedestrians, and weather are always in motion.

A problem-solving agent

- A **problem-solving agent** is an intelligent agent that decides what to do by finding a sequence of actions that leads to the desired goal.

- Unlike reactive agents that respond to current percepts, problem-solving agents use **search** and **planning** to determine actions that are effective over time.
- **A problem-solving agent is a goal-based agent that decides what actions to take by formulating problems and searching for solutions.**
- The agent uses a model of the environment to simulate future states and select action sequences that lead to desirable outcomes.

The Structure of a Problem-Solving Agent

A problem-solving agent has the following structure: •
Goal formulation: Define what it wants to achieve. •

Problem formulation: Translate the goal into a formal problem.

• **Search**: Explore possible actions to find a solution. • **Execution**: Apply the chosen actions in the real world.

Goal Formulation

• The first step is to define the **goal**. A goal is a desired state that the agent wishes to reach. Goals must be:

Well-defined

Achievable

Measurable

• For example, for a delivery robot, the goal might be "Deliver the package to Building B."

Problem Formulation

• Once a goal is set, the agent formulates the problem using the following components:

- **Initial State**: Where the agent starts.
- **Actions (Successor Function)**: What can be done from a given state. • **Transition Model**: Describes the result of performing an action. • **Goal Test**: Determines whether the goal has been reached. • **Path Cost**: Measures the cost of a path (e.g., distance, time).

Search for Solutions

- The agent uses search algorithms to find a path from the initial state to a goal state. This step involves:
- Building a **search tree**
- Exploring nodes (states)
- Evaluating paths using cost or heuristics

Execution

- Finally, the agent executes the selected sequence of actions to achieve the goal. It may re-plan if the environment changes unexpectedly.

Key Components of Problem Formulation in AI

Effective problem-solving in AI is dependent on several important components:

- **Initial State:** This is the starting point of the problem where the AI begins its process. It sets the context and helps identify how the agent will approach the challenge.
- **Action:** At this stage, AI identifies all the possible actions it can take from the initial state. Each action has an impact on how the system moves closer to solving the problem.
- **Transition:** This refers to how the system changes from one state to another after an action is taken. Transition modeling helps show how the agent's actions influence the next steps.
- **Goal Test:** Once an action is taken, AI checks if it has reached its goal. If the goal is achieved, the problem-solving process stops and the solution is considered complete.
- **Cost Function:** This step assigns a numerical value to the cost of achieving the goal. The cost can include resources like time, energy or money and helps decide the most

efficient way to reach the goal.

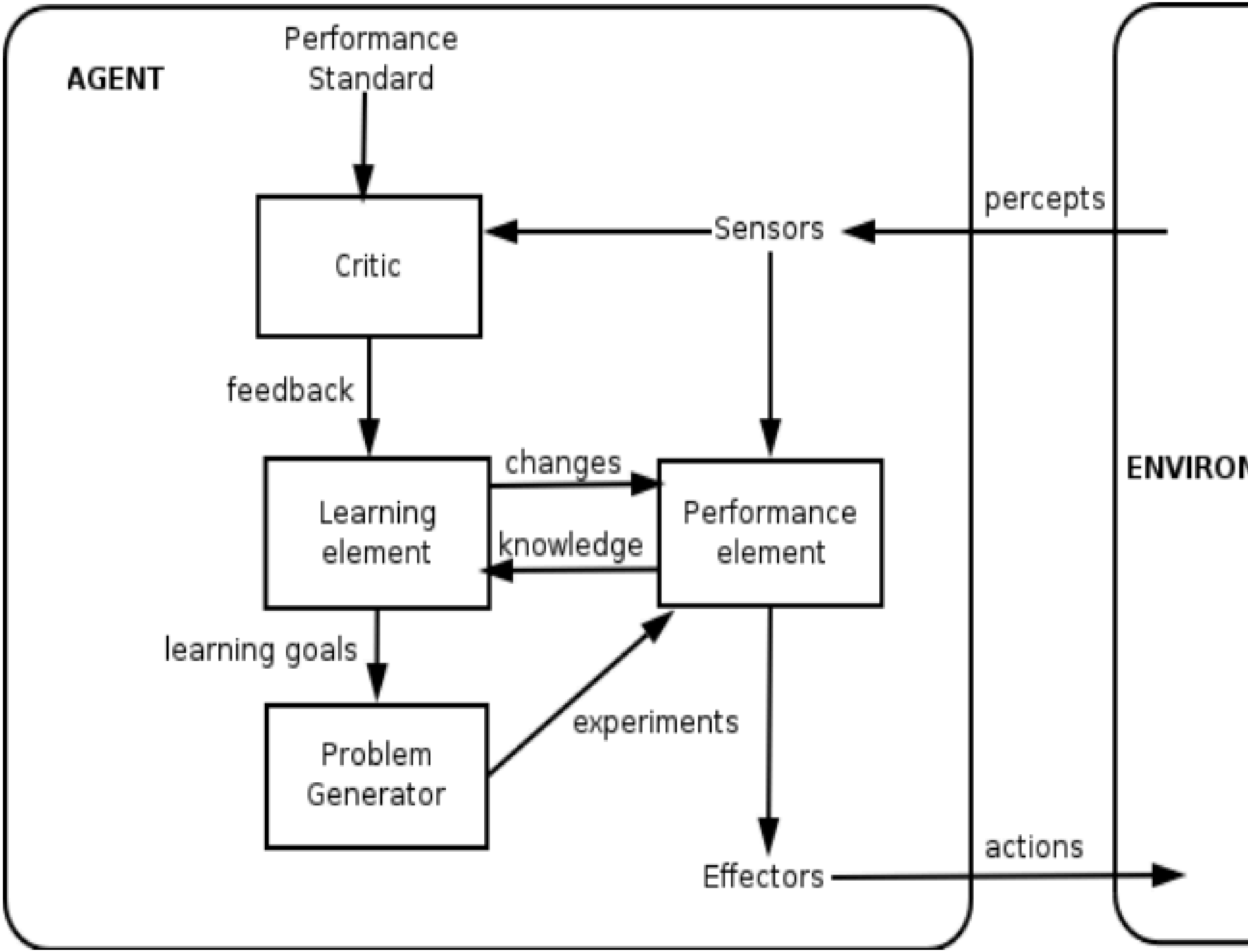
Learning agents

- A learning agent learns about environment model learns about its action consequences and its utility.
- It learns with the feedback from environment.
- Learning enables an agent to adapt new situations without being reprogrammed by humans.

Key characteristics:

- **Adaptive:** Modifies behavior based on experience
- **Autonomous:** Reduces need for explicit programming
- **Generalizes:** Applies learning to new situations

- **Self-improving:** Gets better over time



1. Performance Element
• The "acting" part of the agent
• Takes percepts and decides actions
• Initially uses built-in knowledge/strategies
• Gets improved by learning element

2. Learning Element
• The "learning" part
• Modifies performance element based on feedback
• Responsible for all learning algorithms

3. Critic
• Evaluates how well agent is doing
• Provides feedback (reinforcement signal)
• Tells learning element: "That was good/bad"

4. Problem Generator
• Suggests exploratory actions
• Balances **exploitation** (use known good actions) vs **exploration** (try new actions) • Prevents getting stuck in local optima

Constraint Satisfaction Problem

- A **Constraint Satisfaction Problem** is characterized by:

-- a *set of variables* {x1, x2, .., xn},

-- for each variable xi a *domain* Di with the possible values for that variable, and

-- a set of *constraints*, i.e. relations, that are assumed to hold between the values of the variables

- We will only consider constraints involving one or two variables.

- The constraint satisfaction problem is to find, for each i from 1 to n, a value in Di for xi so that all constraints are satisfied. Means that, we must find a value for each of the variables that satisfies all of the constraints.

Continue..

- A CSP can easily be stated as a first order logic, of the form:

(exist x1)..(exist xn) (D1(x1) & .. Dn(xn) => C1..Cm) • represented as an undirected graph, called *Constraint Graph* where the nodes are the variables and the edges are the binary constraints • Formally, a CSP is defined as a triple where *X* is a set of variables, *D* is a domain of values, and *C* is a set of constraints.

Continue..

- Every constraint is in a pair <t, R>, where *t* is a tuple of variables and *R* is a set of tuples of values; all these tuples having the same number of elements; as a result *R* is a relation.

Constraints

- A constraint is a relation between a **local** collection of variables.
- The constraint restricts the values that these variables can simultaneously have.
- For example, **all-diff(X1, X2, X3)**. This constraint says that X1, X2, and X3 must take on different values. Say that {1,2,3} is the set of

values for each of these variables then: • X1=1, X2=2, X3=3 OK and X1=1, X2=1,X3=3 NO

Crypt arithmetic Problems

- mathematical puzzles where numbers are replaced with alphabets or symbols. • Usually it requires each letter would be replaced by unique digit

Rules

- Digit ranges from 0 to 9 only
- Each variable should have unique value

- Find value for each letter
- The numerical base, unless stated, is 10
- Carry over is 0 or 1
- Numbers must not begin with 0 • After replacing letters by digits, arithmetic operation must be correct

Examples

1. If SEND + MORE = MONEY then find the respective values
2. CROSS + ROADS = DANGER
3. BASE + BALL = GAMES

4. BASIC + LOGIC = PASCAL

5. WRONG+WRONG=RIGHT

6. FATHER+MOTHER=PARENT